Department of Mechanical and Production Engineering

Aarhus University

Exercises

Author:	Noah Rahbek Bigum Hansen
Student ID:	202405538
Course:	Fluid Mechanics — 290211U004
Date:	October 9, 2025

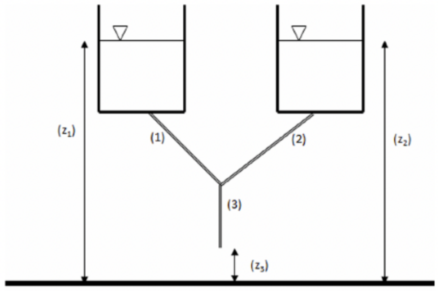
Contents

Lecture 8: Bernoulli Equation Continued

17. September

Problem 6.23

Two water reservoirs at a 30 m elevation each have discharge pipes that are connected at a “tee” junction. One pipe has a 200 mm diameter and the other a 150 mm diameter. The outlet pipe from the “tee” is 300 mm in diameter and discharges to the atmosphere at an elevation of 20 m. Determine the total flow rate.



Assumptions

- Inviscid, incompressible, steady, immiscible, and uniform flow

Derivation

We consider a streamline from reservoir 1 and to the discharge, which we call point 4. Under the assumptions of inviscid and immiscible flow the Bernoulli equation can be applied along this streamline

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + g z_1 = \frac{p_4}{\rho} + \frac{V_4^2}{2} + g z_4.$$

In this case $p_1 = V_1 = p_4 = 0$ and therefore the outlet velocity can be found as

$$\begin{aligned} g z_1 &= \frac{V_4^2}{2} + g z_4 \\ V_4 &= \sqrt{2g\,(z_1 - z_4)} \\ &= \sqrt{2 \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2} \cdot (30\,\mathrm{m} - 20\,\mathrm{m})} \\ &= 14{,}0\,\frac{\mathrm{m}}{\mathrm{s}}. \end{aligned}$$

Knowing the flow speed and the pipe diameter the flow rate can be found using the continuity equation as

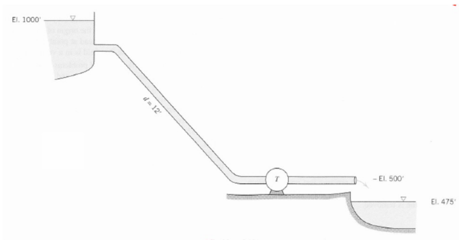
$$Q = A_4 V_4 = \pi \cdot \frac{(300\,\mathrm{mm})^2}{4} \cdot 14{,}0\,\frac{\mathrm{m}}{\mathrm{s}} = 0{,}99\,\frac{\mathrm{m}^3}{\mathrm{s}}.$$

$$\boxed{Q = 0{,}99\,\frac{\mathrm{m}^3}{\mathrm{s}}}$$

RESULT

Problem 6.28

The turbine extracts power from the water flowing from the reservoir. Find the horsepower extracted if the flow through the system is $1000\,\mathrm{ft}^3/\mathrm{s}$. Draw the energy line and the hydraulic grade line.



Assumptions

- Steady, incompressible, inviscid and uniform flow

Derivation

The flow speed through the turbine can be found from the continuity equation

$$Q = A_2 V_2 \implies V_2 = \frac{Q}{A_2} = \frac{1000\,\frac{\mathrm{ft}^3}{\mathrm{s}}}{\pi \cdot (6\,\mathrm{in})^2} = 2{,}7\,\frac{\mathrm{m}}{\mathrm{s}}.$$

From the energy equation we get

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + g z_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2} + g z_2 + T.$$

In this $p_1 = p_2 = V_1 \approx 0$, hence

$$g z_1 = \frac{V_2^2}{2} + g z_2 + T \implies T = g\,(z_1 - z_2) - \frac{V_2^2}{2}.$$

Dividing this by g yields the total head

$$H = z_1 - z_2 - \frac{V_2^2}{2g} = 1000\,\mathrm{ft} - 500\,\mathrm{ft} - \frac{\left(2{,}7\,\frac{\mathrm{m}}{\mathrm{s}}\right)^2}{2 \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2}} = 152\,\mathrm{m}.$$

The formula for finding the hydraulic power from a given head is

$$P_{\mathrm{hyd}} = \rho g Q H = 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3} \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2} \cdot 1000\,\frac{\mathrm{ft}^3}{\mathrm{s}} \cdot 152\,\mathrm{m} = 42{,}2\,\mathrm{kW}.$$

$$\boxed{P_{\mathrm{hyd}} = 42{,}2\,\mathrm{kW}}$$ RESULT

Lecture 9: Dimensional Analysis

22. September 2025

Problem 7.10

A weir is a submerged gate extending across the width of a channel in which water is flowing. The depth of the water upstream of the weir can be used to determine the flow rate. Assume that the volume flow depends on the upstream depth, channel width, and gravity to find the non-dimensional parameters for this situation and determine the dependency of flow rate on the other variables.

Assumptions

- The flow rate depends on the upstream depth, channel width, and gravity.

Derivation

To solve this problem we will utilize the Buckingham-Pi theorem. We let Q denote the volumetric flow rate, g denote gravity, b denote the channel width and h the upstream depth, corresponding to $n = 4$ parameters. We let L and t be our fundamental dimensions and we therefore have $r = 2$ fundamental dimensions. Expressing our 4 parameters based on our two fundamental dimensions we get

$$Q \to \frac{L^3}{t}, \qquad g \to \frac{L}{t^2}, \qquad b \to L, \qquad h \to L.$$

Here we select g and h as our repeating parameters, giving us $m = 2$ repeating parameters. We will have $n - m = 2$ dimensionless groups. Setting up dimensional equations, we obtain

$$\Pi_1 = Q \cdot h^a \cdot g^b \qquad \Longrightarrow \qquad L^0 \cdot t^0 = \frac{L^3}{t} \cdot L^a \cdot \left(\frac{L}{t^2}\right)^b.$$

Now we can sum the exponents to get

$$\begin{aligned} 0 &= 3 + a + b \\ 0 &= -1 - 2b \\ \Longrightarrow b &= -\frac{1}{2} \\ \Longrightarrow a &= -\frac{5}{2}. \end{aligned}$$

Which corresponds to

$$\Pi_1 = Q \cdot h^{-\frac{5}{2}} \cdot g^{-\frac{1}{2}} = \frac{Q}{h^2 \cdot \sqrt{gh}}.$$

We can verify that this is a correct dimensionless group as

$$\left(\frac{L^3}{t}\right) \cdot \left(\frac{1}{L}\right)^2 \cdot \left(\frac{t}{L}\right) = 1.$$

For our second dimensionless group we get

$$\Pi_2 = b \cdot h^a \cdot g^b \qquad \Longrightarrow \qquad L^0 \cdot t^0 = L \cdot L^a \cdot \left(\frac{L}{t^2}\right) b.$$

Summing the exponents yields

$$\begin{aligned} 0 &= 1 + a + b \\ 0 &= -2b \\ \Longrightarrow b &= 0 \end{aligned}$$

$$\implies a = -1.$$

Which corresponds to

$$\Pi_2 = b \cdot h^{-1} \cdot g^0 = \frac{b}{h}.$$

And we can verify this as

$$L \cdot \frac{1}{L} = 1.$$

The functional relationship is $\Pi_1 = f(\Pi_2)$, or

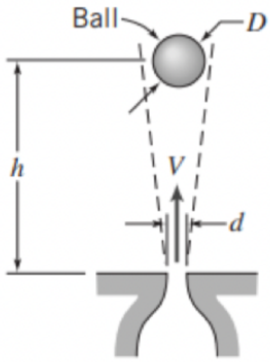
$$\frac{Q}{h^2 \cdot \sqrt{gh}} = f\left(\frac{b}{h}\right).$$

Solving for the flow rate gives

$$\boxed{Q = h^2\sqrt{gh} \cdot f\left(\frac{b}{h}\right)} \qquad \text{RESULT}$$

Problem 7.14

A ball is suspended in an air jet in a stable position as shown on the figure. Experiments show that the equilibrium height of the ball is found to depend on the ball diameter and weight, jet diameter and velocity, and the air density and viscosity. Determine the dimensionless parameters that characterize this experiment.



Assumptions

- The equilibrium height of the ball depends on the ball diameter and weight, the jet diameter and velocity, and the air density and viscosity.

Derivation

We will once again employ the Buckingham-Pi theorem. We let h denote the equilibrium height, d denote the ball diameter, W denote the ball weight, D denote the jet diameter V the jet velocity, ρ the air density and μ the air viscosity. This corresponds to $n = 7$ parameters. We choose mass M, length L and time t as our fundamental dimensions $(r = 3)$ and get

$$h \to L, \qquad d \to L, \qquad W \to \frac{M \cdot L}{t^2}, \qquad D \to L, \qquad V \to \frac{L}{t}, \qquad \rho \to \frac{M}{L^3}, \qquad \mu \to \frac{M}{L \cdot t}.$$

We choose ρ V and d as our repeating parameters giving us $m = 3$. We will have $n - m = 4$ dimensionless groups. Setting up the first dimensional equation we obtain

$$\Pi_1 = h \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = L \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing the exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= a \\ 0 &= -b \\ \Longrightarrow c &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_1 = \frac{h}{d}.$$

Which can be verified as

$$L \cdot \frac{1}{L} = 1.$$

For the second dimensional equation we obtain

$$\Pi_2 = D \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = L \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

This is the same system as before and the exponents will therefore be the same resulting in

$$\Pi_2 = \frac{D}{d}.$$

For the third dimensional equation we obtain

$$\Pi_3 = W \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = \frac{M \cdot L}{t^2} \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= 1 + a \\ 0 &= -2 - b \\ \Longrightarrow a &= -1 \\ b &= -2 \\ c &= -2. \end{aligned}$$

Which corresponds to

$$\Pi_3 = \frac{W}{\rho \cdot V^2 \cdot d^2}.$$

And the fourth dimensional equation gives

$$\Pi_4 = \mu \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = \frac{M \cdot L}{t} \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= 1 + a \\ 0 &= -1 - b \\ \Longrightarrow a &= -1 \\ b &= -1 \\ c &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_4 = \frac{\mu}{\rho \cdot V \cdot d}.$$

$$\boxed{\begin{aligned} \Pi_1 &= \frac{h}{d}, \\ \Pi_2 &= \frac{D}{d}, \\ \Pi_3 &= \frac{W}{\rho \cdot V^2 \cdot d^2}, \\ \Pi_4 &= \frac{\mu}{\rho \cdot V \cdot d}. \end{aligned}}$$

RESULT

Problem 7.18

There are various applications such as freezing of food, combustion of small particles, or some chemical reactions in which a fluid flows through a matrix of particles, called a fluidized bed. The fluid pressure drop is a function of the diameter of the particles d, the flow length of the bed L, the fluid velocity V, fluid density ρ, and fluid viscosity μ. Use dimensional analysis to find the functional dependence of pressure drop on the other variables.

Assumptions

- The fluid pressure drop depends on the diameter of the particles, the flow length, the fluid velocity, the fluid density and the fluid viscosity

Derivation

We let the fluid pressure drop be termed Δp and in total we have $n = 6$ parameters. We choose mass M, length L and time t as our fundamental dimensions corresponding to $r = 3$. Our parameters expressed in terms of these are

$$\Delta p \to \frac{M}{L \cdot t^2}, \qquad d \to L, \qquad L \to L, \qquad V \to \frac{L}{t}, \qquad \rho \to \frac{M}{L^3}, \qquad \mu \to \frac{M}{L \cdot t}.$$

We choose L, V, and ρ as our repeating parameters giving $m = 3$. This requires $n - m = 3$ dimensionless groups. Setting up the first of these gives

$$\Pi_1 = \Delta p \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = \frac{M}{L \cdot t^2} \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing the exponents gives

$$\begin{aligned} 0 &= -1 + a + b - 3c \\ 0 &= 1 + c \\ 0 &= -2 - b \\ \Longrightarrow c &= -1 \\ b &= -2 \\ a &= 0. \end{aligned}$$

This corresponds to

$$\Pi_1 = \frac{\Delta p}{V^2 \rho}.$$

The second dimensionless group becomes

$$\Pi_2 = d \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = L \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 + a + b - 3c \\ 0 &= c \\ 0 &= -b \\ \Longrightarrow a &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_2 = \frac{d}{L}.$$

The third dimensionless group is

$$\Pi_3 = \mu \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = \frac{M}{L \cdot t} \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= -1 + a + b - 3c \\ 0 &= 1 + c \\ 0 &= -1 - b \\ \Longrightarrow b &= -1 \\ c &= -1 \\ a &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_3 = \frac{\mu}{L \cdot V \cdot \rho}.$$

The functional relationship is therefore

$$\frac{\Delta p}{\rho V^2} = f\left(\frac{d}{L}, \frac{\mu}{L \cdot V \cdot \rho}\right).$$

Here $\frac{\Delta p}{\rho V^2}$ is the pressure coefficient and $\frac{\mu}{L \cdot V \cdot \rho} = \frac{1}{\mathrm{Re}}$.

$$\boxed{\frac{\Delta p}{\rho V^2} = f\left(\frac{d}{L}, \frac{\mu}{L \cdot V \cdot \rho}\right)}$$ RESULT

Lecture 10: Internal Flows: Fully Developed Laminar Flows

24. September 2025

Problem 8.42

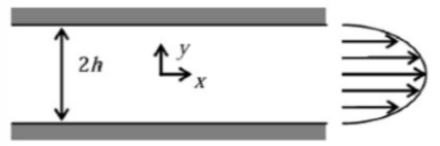
In the laminar flow of an oil of viscosity 1 Pa s, the velocity at the center of a 0,3 m pipe is 4,5 $\frac{\text{m}}{\text{s}}$ and the velocity distribution is parabolic. Calculate the shear stress at the pipe wall and within the fluid 75 mm from the pipe wall.

Problem 8.13

When a horizontal laminar flow occurs between two parallel plates of infinite extent 0,3 m apart, the velocity at the midpoint between the plates is 2,7 $\frac{\text{m}}{\text{s}}$. Calculate (a) the flow rate through a section 0,9 m wide, (b) the velocity gradient at the surface of the plate, (c) the wall shearing stress if the fluid has viscosity 1,44 Pa s, (d) the pressure drop in each 30 m along the flow.

Problem 8.6

A fully developed and laminar flow of oil occurs between parallel plates. The pressure gradient creating the flow is 1 25 $\frac{\text{kPa}}{\text{m}}$ of length and the channel half-width is 1,5 mm. Calculate the magnitude and length direction of the wall shear stress at the upper plate surface. Find the volume flow rate through the channel. The viscosity is 0,50 N s/m^2.



Lecture 11: Internal Flows: Flows in Pipes and Ducts

29. September 2025

Problem 8.20

For a turbulent flow of a fluid in 0,6 m diameter pipe, the velocity 0,15 m from the wall is 2,7 $\frac{\text{m}}{\text{s}}$. Estimate the wall shear stress using the 1/7$^{\text{th}}$ expression for the velocity profile.

Problem 8.24

A pipe runs for an elevation of 45 m to an elevation of 115 m. The inlet pressure is 8,5 MPa and the head loss is 6,9 $\frac{\text{kJ}}{\text{kg}}$. Calculate the outlet pressure for (a) the inlet at the 45 m elevation and (b) the inlet at the 115 m elevation.

Problem 8.28

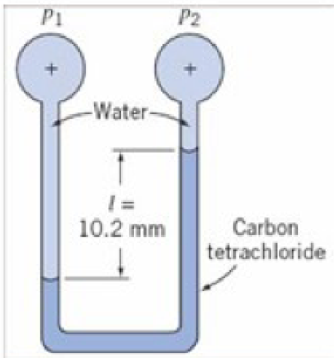
Water flows in a smooth 0,2 m diameter pipeline that is 65 m long. The Reynolds number is 10^6. Determine the flow rate and pressure drop. After many years of use, minerals deposited on the pipe cause the same pressure drop to produce only one-half the original flow rate. Estimate the size of the relative roughness elements for the pipe.

Lecture 12: Recap

1. October 2025

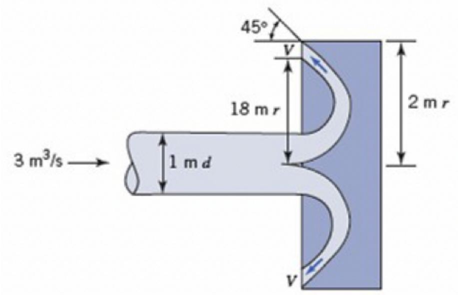
Problem 3.11

Consider the two-fluid manometer shown. Calculate the applied pressure difference.



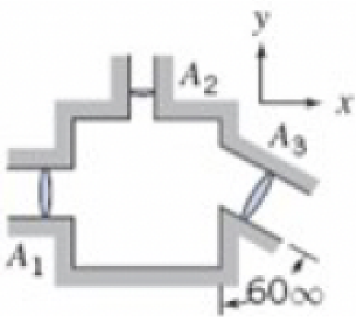
Problem 4.10

Find V for this mushroom cap on a pipeline.



Problem 4.4

Fluid with 1040 $\frac{\text{kg}}{\text{m}^3}$ density is flowing steadily through the rectangular box shown. Given $A_1 = 0{,}046\,\text{m}^2$, $A_2 = 0{,}009\,\text{m}^2$, $A_3 = 0{,}056\,\text{m}^2$, $\boldsymbol{V}_1 = \left(3\,\frac{\text{m}}{\text{s}}\right)\hat{\imath}$, and $\boldsymbol{V}_2 = \left(6\,\frac{\text{m}}{\text{s}}\right)\hat{\jmath}$. Determine the velocity $\boldsymbol{V}_3$.



Problem 5.19

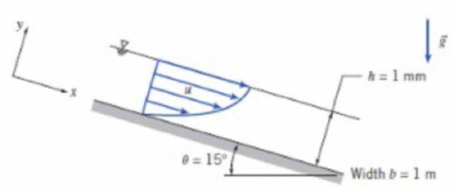
A velocity field is given by $\boldsymbol{V} = 2\hat{\imath} - 4x\hat{\jmath}\,\frac{\text{m}}{\text{s}}$. Determine an equation for the streamline. Calculate the vorticity of the flow.

Problem 5.6

The x component of velocity for a flow field is given as $u = Ax^2y^2$ where $A = 0{,}3\,\mathrm{m}^{-3}\,\mathrm{s}^{-1}$ and x and y are in meters. Determine the y component of velocity for a steady incompressible flow. Determine the equation of the streamline and plot the streamline that goes through the point $(x, y) = (1, 4)$.

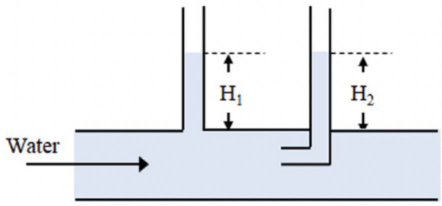
Problem 5.25

A liquid film flows on a horizontal surface due to a constant shear stress on the top surface. The liquid film is thin, the flow is fully developed, and the thickness is constant in the flow direction. Determine the velocity profile $u(x)$ and the pressure gradient $\frac{\mathrm{d}p}{\mathrm{d}x}$.



Problem 6.4

Water flows in a pipe at a velocity of $3\,\frac{\mathrm{m}}{\mathrm{s}}$ and a pressure of 120 kPa absolute. Two tubes are placed in the tube as shown in the figure. Determine the height of the water in each column and the stagnation pressure in kPa.



Problem 7.22

A flat plate 1,5 m long and 0,3 m wide is towed at $3\,\frac{\mathrm{m}}{\mathrm{s}}$ in a towing basin containing water at 20 °C. The measured drag force is 14 N. Calculate the dimensions of a similar plate that will yield dynamically similar conditions in an airstream at a velocity of $18\,\frac{\mathrm{m}}{\mathrm{s}}$, 101 kPa and 15 °C. Determine the expected drag force on the plate in the air stream.

Problem 8.17

A fluid of specific gravity 0,9 flows at a Reynolds number of 1500 in a 0,3 m pipeline. The velocity 50 mm from the wall is $3\,\frac{\mathrm{m}}{\mathrm{s}}$. Calculate the flow rate and the velocity gradient at the wall.